# Distinguishing Coherent Crosstalk from Calibration Drift via Pauli-Transfer Signatures and Quantum Edge Detection

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***Abstract*—Multi-tenant quantum processors expose a pulse-level attack surface in which coherent crosstalk can mimic benign calibration drift at matched average gate infidelity. We present a structural verification and detection framework based on the residual Pauli transfer matrix (PTM). We prove that products of local unital, trace-preserving channels cannot mix weight-1 and weight-2 Pauli operators, even under coherent drift of arbitrary strength or axis. Building on established PTM-to-Hamiltonian relations, we prove that, for any specified qubit pair in an arbitrarily large system, the first-order map from its nine interaction coefficients to the antisymmetric cross-weight feature is an isometry up to scale. Thus, every interaction direction is locally observable near the identity. We also exhibit cancellation of this feature under large local rotations and prove that the full cross-weight norm is invariant under local unitary composition, providing a structural countermeasure. For calibrated small local rotations, shared randomized-Pauli measurements support periodic verification using the evaluated antisymmetric detector. In simulation, $16{,}384$ randomized settings yield a detection threshold of $\lambda_{\min} \approx 0.13$ at a $5\%$ calibrated false-positive target, with approximately $M^{-1/2}$ scaling in the number of settings $M$. Further experiments characterize detection under relaxation, depolarization, native ZZ fluctuations, and time-dependent pulse dynamics. Finally, an FRQI encoding with Pauli-graph quantum Hadamard edge detection reproduces the classical structural edge score to numerical precision.**

***Index Terms*—Quantum hardware security; Coherent crosstalk; Calibration drift; Pauli transfer matrix; Randomized-Pauli shadows; Edge detection; FRQI**

## I. Introduction

Quantum processors are increasingly offered as shared, cloud-scheduled systems. In these environments, several users may access nearby qubits, and pulse-level control can allow disturbances on one qubit to affect another. Such disturbances may arise from cross-resonance leakage, spectator-conditioned drives, imperfect echo cancellation, or unwanted always-on coupling [1]–[4]. These effects create a hardware-security concern because a weak coherent interaction can be introduced during a victim gate without producing an obvious failure.

The main challenge is that coherent crosstalk can resemble ordinary calibration drift. A local over-rotation and a two-qubit interaction can be chosen to produce the same average gate infidelity. Standard scalar diagnostics, including average fidelity and randomized benchmarking error, can then reveal that the gate is inaccurate but cannot determine whether the cause is local drift or a nonlocal interaction. This ambiguity matters because the two cases require different responses. Local drift may require recalibration, while an unexpected two-qubit interaction may indicate control leakage, faulty isolation, or an injected pulse-level disturbance.

Prior work has studied the characterization, mitigation, and calibration of crosstalk in superconducting processors [1], [3], and off-diagonal PTM entries have been related to local and two-qubit Hamiltonian coefficients [5] and estimated from randomized product-Pauli data [6]. Recent security studies have also shown that structured pulse-level perturbations can alter a victim computation while remaining difficult to identify through conventional performance metrics [2]. These results do not, however, provide a security criterion for separating local coherent drift from coherent crosstalk when their error magnitudes are matched; we build on these characterization tools to formulate one. The Pauli transfer matrix describes how a channel transforms one Pauli operator into another [7], [8], and randomized-Pauli and classical-shadow methods estimate many of its entries from one shared dataset [9], [10]. The remaining question is which part of it reliably distinguishes local drift from nonlocal coherent interaction.

We show that the relevant information lies in the signed antisymmetric coupling between weight-1 and weight-2 Pauli sectors. Products of local unital, trace-preserving errors, including coherent drift of arbitrary strength, axis, and location, do not create such mixing, whereas a genuine two-qubit coherent interaction does at first order. This defines a signed Pauli-transfer signature for direction-resolved detection relative to an attack-free calibration baseline.

We also study a quantum-native readout. Because neighboring PTM indices need not represent related Pauli operators, we define a Pauli graph whose edges connect labels that differ in one tensor factor, encode the PTM-derived magnitude pattern as a flexible representation of quantum images (FRQI) state [11], and measure its variation over this graph with quantum Hadamard edge detection (QHED) [12], [13]. The signed PTM vector remains the detector.

This work addresses the following questions:

**RQ1.** Can coherent two-qubit crosstalk be distinguished from local calibration drift when both produce the same average gate infidelity?

**RQ2.** Does the proposed cross-weight Pauli-transfer signature capture every two-qubit Pauli interaction direction, or can some interactions remain invisible?

**RQ3.** Can this signature be estimated with finite randomized measurements and remain effective under relaxation,

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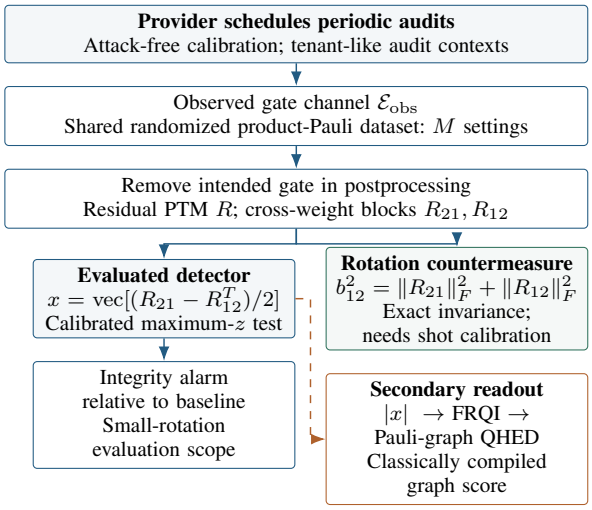



Fig. 1. Overview of the framework. One shared randomized-Pauli dataset yields the residual PTM and the signed cross-weight vector $x$, which is exactly zero for product local unital errors (Proposition 1) and first order in any two-qubit Pauli generator (Proposition 2). The calibrated direction-resolved test on $x$ carries the security decision within the small-rotation scope; the full-block norm $b_{12}$ is invariant under local rotations (Proposition 3), and the FRQI and Pauli-graph QHED branch is a quantum-native readout of the feature's structural pattern.

depolarizing noise, native coupling fluctuations, and time-dependent pulse dynamics?

**RQ4.** Can the resulting PTM feature be represented and measured through an FRQI and Pauli-graph quantum edge-detection circuit?

**Contributions:** In answer to these questions, this paper makes four contributions, and we distinguish their levels of evidence explicitly. First, it proves an exact algebraic result: the weight-1 to weight-2 block of the residual PTM is identically zero for every product of local unital, trace-preserving error channels, and any two-qubit Pauli generator populates this block at first order (Section IV). Second, building on the established first-order PTM-to-Hamiltonian relation [5], it proves that the map from the nine interaction coefficients into this block is an isometry up to scale for every system size, and it gives a full-block invariant for large local rotations that cancel the antisymmetric projection. Third, it specifies a single operational detector, estimated from one shared randomized-Pauli measurement dataset and calibrated on attack-free data, and evaluates it empirically under realistic noise, finite measurement budgets, fluctuating native ZZ coupling, and time-ordered pulse dynamics (Sections V and VI). Fourth, it constructs a Pauli-graph QHED circuit that measures the structural pattern of the PTM feature from an FRQI encoding and verifies the circuit against the classical score (Section VII).

The PTM discriminator and the calibrated hypothesis test are the primary security contribution; the FRQI and QHED construction is a secondary representation result that improves neither detection accuracy nor computational complexity. Figure 1 summarizes the pipeline.

The method detects unexpected nonlocal coherent structure relative to a calibrated baseline; it does not infer malicious intent, and all detection results are hardware-motivated simulations at three qubits. Section VIII states the detection boundary.

## II. Background and Related Work

### A. Crosstalk characterization

Crosstalk is a dominant error source in superconducting processors, and a substantial literature addresses its detection and modeling. Sarovar et al. [1] formalize crosstalk errors as violations of locality and independence assumptions and propose protocols that detect their presence. Simultaneous gate set tomography distinguishes crosstalk mechanisms experimentally on trapped-ion and transmon hardware [4]. Recent characterization work shows that benign transmon hardware carries persistent parasitic ZZ coupling and two-level-system mediated interactions that can be learned with a few parameters per qubit pair [3]. These studies target device bring-up and calibration rather than the security question of whether an observed coherent error is local or nonlocal at matched magnitude, which is the gap this paper fills.

### B. Pulse-level attacks on shared hardware

Security analyses of multi-tenant quantum computing have demonstrated that pulse-level access enables crosstalk-based attacks. Prior work simulates cross-resonance pulse injections that corrupt a victim circuit and characterizes stealthy variants that remain below conventional detection thresholds [2]. These studies establish the attack surface but leave open how a defender can attribute an observed error to an injected interaction rather than to drift. As a countermeasure, that line of work proposes canary circuits: simple probe circuits executed periodically on idle qubits and flagged when their output deviates from a pre-calibrated baseline [2]. Canaries inherit the scalar limitation, since a deviation shows that the device has changed but not whether the cause is drift or an injected interaction, and an attack matched to the calibration budget can hide inside the baseline tolerance. The protocol in this paper can be read as a structural canary that addresses this gap: the randomized-Pauli audit circuits play the canary role, while the cross-weight signature replaces the scalar baseline comparison with a mechanism-resolved and direction-resolved test.

### C. Channel estimation and coherence of noise

The Pauli transfer matrix is a standard representation of quantum channels [7], [8]. Classical-shadow and randomized-Pauli methods estimate many channel properties from one shared dataset with cost that depends on operator weight rather than system size [9], [10]. Kaufmann et al. [5] relate off-diagonal PTM entries to all single- and two-qubit coherent rotation coefficients and estimate them on superconducting hardware; our $C$ and $L$ (Section IV) use that relation. Wallman et al. [14] estimate how coherent a noise

process is, which is a scalar property. Our contribution is not a new estimation protocol but a security test on a specific PTM block whose occupancy separates local from nonlocal coherent errors, which no scalar coherence measure can do.

### D. Quantum image processing

FRQI encodes an image into rotation angles of a color qubit entangled with an address register [11], and QHED measures intensity differences between neighboring pixels with a Hadamard transform [12], [13]. We replace pixel adjacency with a Pauli-graph adjacency for operator-indexed channel data.

## III. Threat Model

### A. System model

Consider an $n$-qubit processor containing a victim qubit $v$ and one or more spectator qubits. The intended operation on the monitored subsystem is the unitary channel $\mathcal{U}_{\mathrm{id}}(\rho) = U_{\mathrm{id}}\rho U_{\mathrm{id}}^{\dagger}$. The actual implementation is a noisy channel $\mathcal{E}_{\mathrm{obs}}$. The defender wishes to determine whether the difference between $\mathcal{E}_{\mathrm{obs}}$ and $\mathcal{U}_{\mathrm{id}}$ is consistent with the calibrated device or contains an additional coherent two-qubit interaction.

### B. Calibrated null model

Let $\mathfrak{N}$ denote the family of channels expected during normal device operation: depolarizing noise, dephasing, amplitude damping, thermal relaxation, local coherent miscalibration of the form $U_{\mathrm{loc}} = \bigotimes_q e^{-i\epsilon_q P_q/2}$, and calibrated always-on parasitic coupling. The calibrated hardware baseline may contain persistent two-qubit ZZ coupling and TLS-mediated interactions; such terms can be estimated using pairwise characterization methods such as JAZZ-type amplification experiments [3]. Under the alternative hypothesis below, local rotations composed with an attack satisfy $|\epsilon_q| \le \epsilon_{\max}$ with $2[\sin(\epsilon_{\max}/2) + \sin\epsilon_{\max}] < 1$; $\epsilon_{\max} = 0.3$ rad suffices and exceeds the $0.15$ rad drift of the finite-shot studies.

The exact structural-zero result of Section IV applies to products of local unital, trace-preserving channels. The complete hardware null need not satisfy this product assumption, because amplitude damping is nonunital and native ZZ is a two-qubit interaction. Operational detection is therefore performed relative to the measured calibration distribution, not relative to the mathematical zero alone.

### C. Attack model

The monitored attack family consists of coherent two-qubit interactions applied during the victim gate. A general interaction on spectator $s$ and victim $v$ is

$$H_{\mathrm{int}} = \sum_{a,b\in\{X,Y,Z\}} h_{ab}\,\sigma_a^{(s)} \otimes \sigma_b^{(v)}, \tag{1}$$

where the nine real coefficients $h_{ab}$ specify the interaction direction. The corresponding unitary attack is $U_{\mathcal{A}}(\lambda) = e^{-i\lambda H_{\mathrm{int}}/2}$ with channel $\mathcal{A}_\lambda(\rho) = U_{\mathcal{A}}\rho U_{\mathcal{A}}^{\dagger}$. Here $\lambda$ is the dimensionless interaction angle accumulated over the monitored gate: for a constant physical coupling of strength $J$ entering the Hamiltonian as $JH_{\mathrm{int}}/2$ over gate duration $t_g$, $\lambda = Jt_g$, and for the time-dependent envelopes of Section VI-F it is the corresponding integrated angle. The evaluated attack families include ZX, YX, and excess ZZ interactions, together with pulse-level realizations such as detuned cross-resonance leakage and spectator-conditioned drives.

Against an adaptive adversary who knows the detector, this class is motivated by pulse-level cross-resonance access, which naturally produces coherent spectator-conditioned terms that change the channel at first order [2]. Stochastic correlated attacks evade the antisymmetric signature (Section VIII), and excess ZZ hidden inside the native fluctuation band is capped at that band (Section VI-E).

### D. Defender capability

The defender can schedule diagnostic circuits containing the victim gate and collect randomized Pauli preparation and measurement data under controllable conditions $c \in \{\mathrm{idle}, \mathrm{driven}, \mathrm{calibration\ time}, \mathrm{observation\ time}\}$. Scheduling audits and controlling the spectator condition require provider-side access, so the primary deployment is a provider-run integrity monitor. A tenant without scheduling privileges can still run the passive single-condition variant on its own gates, which retains the structural discriminator but not the context-contrast handle. The deployment assumes an attack-free calibration window and audits that the adversary cannot distinguish from tenant jobs; an adversary that recognizes audit windows and stays silent evades observation (Section VIII).

No privileged access to analog control electronics is required: Clifford diagnostic gates reduce removal of the intended unitary to classical Pauli relabeling, and for a non-Clifford gate the residual PTM remains estimable by linearity at a post-processing cost set by the number of resulting Pauli terms.

### E. Hypothesis test

The residual error channel is $\mathcal{E}_{\mathrm{err}} = \mathcal{U}_{\mathrm{id}}^{\dagger}\circ\mathcal{E}_{\mathrm{obs}}$. The defender tests

$$\begin{aligned} H_0 &: \ \mathcal{E}_{\mathrm{err}} \in \mathfrak{N}, \\ H_1 &: \ \mathcal{E}_{\mathrm{err}} = \mathcal{D}\circ\mathcal{M}_{\epsilon'}\circ\mathcal{A}_\lambda\circ\mathcal{M}_{\epsilon}, \\ &\quad\ |\epsilon_q|, |\epsilon_q'| \le \epsilon_{\max}, \quad \lambda > \lambda_{\min}, \end{aligned} \tag{2}$$

at a controlled false-positive rate, where $\mathcal{M}_\epsilon$ and $\mathcal{M}_{\epsilon'}$ are local rotations before and after the interaction and $\mathcal{D}$ is the remaining calibrated noise in $\mathfrak{N}$. The angular bound is necessary: a large local rotation, such as a spectator $\pi$ rotation, can cancel the antisymmetric signature (Section IV-F). Thresholds are selected from held-out calibration data, and Table I states which attack classes each handle addresses.

## IV. The Structural Pauli-Transfer Signature

### A. Pauli transfer representation

The $n$-qubit Pauli basis is $\mathcal{P}_n = \{P_\alpha = \bigotimes_q P_{\alpha_q} : P_{\alpha_q} \in \{I, X, Y, Z\}\}$ with $\mathrm{Tr}(P_\alpha P_\beta) = 2^n\delta_{\alpha\beta}$. The weight $\mathrm{wt}(P_\alpha)$

TABLE I

DETECTABILITY BOUNDARY OF THE PROPOSED PROTOCOL. ✓ INDICATES DIRECT SENSITIVITY, × INDICATES THAT THE ATTACK CAN EVADE THE CORRESPONDING TEST, AND ○ INDICATES PARTIAL SENSITIVITY. CROSS-WEIGHT REFERS TO THE EVALUATED ANTISYMMETRIC STATISTIC $x$; THE FULL-BLOCK NORM OF PROPOSITION 3 RESTORES SENSITIVITY IN THE $\pi$-ROTATION ROW.

| Attack class | Cross-weight | Temporal | Context |
|---|---|---|---|
| Coherent two-qubit interaction | ✓ | ✓ | ✓ |
| Interaction + local $\pi$ rotation | × | ○ | ○ |
| Local coherent drift | 0 (Prop. 1) | ○ | 0 |
| Incoherent correlated noise | × | ○ | ○ |
| Coherent interaction of weight $> k$ | × | ○ | ○ |
| Excess ZZ inside native band | ○ | ○ | ○ |
| Attack present during calibration | × | × | ○ |

counts the non-identity tensor factors; for example $\mathrm{wt}(IIX) = 1$ and $\mathrm{wt}(IZX) = 2$. We monitor the low-weight sector $\mathcal{L}_k = \{P : \mathrm{wt}(P) \le k\}$ of size $K = \sum_{w=0}^{k} 3^w \binom{n}{w}$; throughout the numerics $n = 3$ and $k = 2$, so $K = 37$. For fixed $k$, $K$ grows polynomially in $n$, and the low-weight block is estimable by classical shadows at polynomial cost [9], [10].

The PTM of the residual channel is

$$R_{\alpha\beta} = 2^{-n} \operatorname{Tr}[P_\alpha \, \mathcal{E}_{\mathrm{err}}(P_\beta)] \,, \tag{3}$$

so that $\mathcal{E}_{\mathrm{err}}(P_\beta) = \sum_\alpha R_{\alpha\beta} P_\alpha$. The PTM is real for physical channels, equals the identity for a perfect gate, and has an orthogonal traceless part for unitary channels [7], [8].

For a coherent error $U_\lambda = e^{-i\lambda G/2}$ generated by a Hermitian $G$, expanding $U_\lambda P_\beta U_\lambda^\dagger = P_\beta - \frac{i\lambda}{2}[G, P_\beta] + O(\lambda^2)$ gives $R = I + \lambda C + O(\lambda^2)$, where $C_{\alpha\beta} = 2^{-n}\operatorname{Tr}[P_\alpha(-\frac{i}{2})[G, P_\beta]]$ satisfies $C^T = -C$. Coherent errors therefore appear, at first order, in the antisymmetric part of the PTM. This motivates the signed component

$$A_{\alpha\beta} = \tfrac{1}{2} \left( R_{\alpha\beta} - R_{\beta\alpha} \right) , \tag{4}$$

which retains the sign of each coherent Pauli transition and hence the direction of the underlying Hamiltonian interaction.

### B. The cross-weight signature vector

Define the oriented index set $\mathcal{I}_{12} = \{(\alpha, \beta) : \mathrm{wt}(P_\alpha) = 2, \ \mathrm{wt}(P_\beta) = 1\}$ and the signed cross-weight block $A^{12}_{\alpha\beta} = A_{\alpha\beta}$ for $(\alpha, \beta) \in \mathcal{I}_{12}$. The structural signature is the vector

$$x = \mathrm{vec}\big(A^{12}\big) \in \mathbb{R}^{K_1 K_2}, \tag{5}$$

where $K_1 = 3n$ and $K_2 = 9\binom{n}{2}$ count the weight-1 and weight-2 Pauli strings. For $n = 3$, $K_1 = 9$, $K_2 = 27$, and $x \in \mathbb{R}^{243}$. Each coordinate of $x$ is one signed coherent transition from a weight-1 input Pauli to a weight-2 output Pauli.

Both ingredients of this definition are essential. Restricting to cross-weight entries excludes local unital product errors (Proposition 1 below), while antisymmetrization selects the first-order coherent component and suppresses symmetric stochastic mixing; unlike the cross-weight restriction, it is not invariant under large local rotations (Section IV-F). Nonunital product noise, such as amplitude damping, can still create raw cross-weight leakage; this residual contribution is handled operationally through baseline calibration (Sections V and VI-B).

### C. Structural zero for local unital errors

**Proposition 1** (Local errors have zero cross-weight mass)**.** *Let $\mathcal{E} = \bigotimes_q \mathcal{E}_q$ be any tensor product of single-qubit channels, each trace preserving and unital. Then every PTM entry with $\mathrm{wt}(P_\alpha) \neq \mathrm{wt}(P_\beta)$ vanishes, and in particular $x = 0$. This holds for local coherent drift $e^{-i\epsilon P_q/2}$ of any strength $\epsilon$, on any qubit, about any axis.*

*Proof.* The PTM of a product channel factorizes as $R_{\alpha\beta} = \prod_q R^{(q)}_{\alpha_q\beta_q}$. If $\mathrm{wt}(P_\alpha) \neq \mathrm{wt}(P_\beta)$, there is a site $q$ where exactly one of $\alpha_q, \beta_q$ is the identity. At that site the factor is either $R^{(q)}_{I,P} = \frac{1}{2}\mathrm{Tr}[\mathcal{E}_q(P)] = 0$, because the channel is trace preserving and $P$ is traceless, or $R^{(q)}_{P,I} = \frac{1}{2}\mathrm{Tr}[P\,\mathcal{E}_q(I)] = 0$, because the channel is unital. Either way the product vanishes. □

This result is exact rather than perturbative: local coherent drift is not merely small in the monitored block, it is algebraically forbidden from entering it.

### D. First-order response to two-qubit interactions

**Proposition 2** (Two-qubit generators are visible at first order)**.** *Let $\mathcal{E}_{\mathrm{err}}(\rho) = e^{-i\lambda G/2}\rho\, e^{i\lambda G/2}$ with $G = P_s \otimes P_v$ a two-qubit Pauli. Then $\|x\|_2 = \Theta(|\lambda|)$ as $\lambda \to 0$.*

*Proof sketch.* Choose a weight-1 Pauli $Q$ on the victim with $\{Q, P_v\} = 0$. Then $[G, Q] = P_s \otimes [P_v, Q]$ is proportional to a weight-2 Pauli, giving a nonzero entry of the antisymmetric generator $C$ in the weight-1 to weight-2 block, hence mass proportional to $\lambda$ at first order. □

As a concrete example, take $G = Z_s X_v$ and input $P_\beta = I_s Y_v$. Since $[Z_s X_v, I_s Y_v] = 2i Z_s Z_v$, conjugation gives $\cos(\lambda)\, I_s Y_v + \sin(\lambda)\, Z_s Z_v$: the weight-1 input acquires a weight-2 component whose PTM entry scales as $\sin(\lambda) = \lambda + O(\lambda^3)$.

### E. No monitored interaction direction is invisible

Proposition 2 covers a single Pauli product, but a physical interaction is the general Hamiltonian of Eq. (1). Collecting the nine coefficients into $h \in \mathbb{R}^9$, the first-order signature is linear,

$$x = \lambda\, L h + O(\lambda^2), \tag{6}$$

with $L : \mathbb{R}^9 \to \mathbb{R}^{K_1 K_2}$ determined by the Pauli commutators. Two failure modes would undermine detection: a nonzero interaction in the null space of $L$ would be invisible, and two interaction directions could cancel.

Neither occurs, for any system size.

**Lemma 1** (Pair-local isometry)**.** *For any $n \ge 2$, $k \ge 2$, and any fixed pair $(s, v)$, $L^T L = 4 I_9$. Thus $\mathrm{rank}(L) = 9$, all nine singular values equal 2, and the nonzero entries of $L$ are 36 values $\pm 1$ in rows whose Paulis are supported on the pair.*

*Proof.* For $G_{ab} = \sigma_a^{(s)}\sigma_b^{(v)}$, exactly four weight-1 inputs anticommute with $G_{ab}$, two on each qubit of the pair, and the commutator maps each to $\pm 1$ times a weight-2 Pauli on the pair; inputs on other qubits commute with $G_{ab}$. Each input-output coordinate identifies $(a, b)$, so distinct columns have disjoint supports of squared norm 4. The normalized trace in $C$ removes identity factors on other qubits, so larger registers only add zero rows. □

The same construction applied to the six local generators $\sigma_a^{(s)} \otimes I$ and $I \otimes \sigma_b^{(v)}$ yields exactly zero columns, consistent with Proposition 1, and Pauli-algebra checks at $n = 2, \ldots, 5$ reproduce the lemma exactly. Consequently, every two-qubit interaction direction produces cross-weight mass, $\|x\|_2 = 2|\lambda|\,\|h\|_2 + O(\lambda^2)$, no two directions cancel, and $x$ faithfully encodes the nine-parameter interaction tensor near the identity; Section IV-F treats composition with local rotations. This answers RQ2 and justifies the direction-resolved detector of Section V: distinct interactions map to distinct, non-cancelling directions in $x$, so a per-coordinate statistic can separate an attack direction from a different native-coupling direction.

### F. Local-rotation cancellation and full-block invariance

Write $R_{rs}$ for the block of $R$ with weight-$r$ rows and weight-$s$ columns, so that $x = \mathrm{vec}(A_{21}^-)$ with $A_{21}^{\mp} = (R_{21} \mp R_{12}^T)/2$. For $V_\epsilon = e^{-i\epsilon X_s/2}$ composed in either order with $U_{\mathcal{A}} = e^{-i\lambda Z_s X_v/2}$,

$$\|A_{21}^-\|_F = 2|\sin\lambda\cos\tfrac{\epsilon}{2}|, \quad \|A_{21}^+\|_F = 2|\sin\lambda\sin\tfrac{\epsilon}{2}|, \tag{7}$$

and the same holds for YX and ZZ. At $\epsilon = \pi$, $x = 0$ for every $\lambda$, although the channel remains nonlocal when $\sin\lambda \neq 0$. Halving $\|x\|_2$ already requires $|\epsilon| \geq 2\pi/3$, a spectator error with single-qubit average infidelity $(1 - \cos\epsilon)/3 \geq 0.5$ that is visible in the same-weight block $R_{11}$ and in routine calibration.

**Proposition 3** (Full-block invariance)**.** *Let* $b_{12}(R)^2 = \|R_{21}\|_F^2 + \|R_{12}\|_F^2$. *For product-unitary channels* $\mathcal{V}_{\mathrm{in}}, \mathcal{V}_{\mathrm{out}}$ *and any channel* $\mathcal{E}$, *the PTM* $R'$ *of* $\mathcal{V}_{\mathrm{out}} \circ \mathcal{E} \circ \mathcal{V}_{\mathrm{in}}$ *satisfies* $b_{12}(R') = b_{12}(R)$.

*Proof.* Product unitaries preserve Pauli weight and act orthogonally on each weight sector, so $R'_{21} = O_2^{\mathrm{out}} R_{21} O_1^{\mathrm{in}}$ and $R'_{12} = O_1^{\mathrm{out}} R_{12} O_2^{\mathrm{in}}$. □

In Eq. (7), $b_{12} = 2\sqrt{2}|\sin\lambda|$ for every $\epsilon$. This full-block countermeasure is not evaluated at finite shots; all detection results use $x$. For small rotations, a first-order expansion with the weight-preserving orthogonal PTMs $O$ and $O'$ of $\mathcal{M}_\epsilon$ and $\mathcal{M}_{\epsilon'}$ gives $\|x\|_2 \geq 2(1 - \eta - \eta')|\lambda|\,\|h\|_2 - O(\lambda^2)$, where $\eta = (\|O_1 - I\|_2 + \|O_2 - I\|_2)/2 \leq \sin(\epsilon_{\max}/2) + \sin\epsilon_{\max}$ independently of $n$ and $\eta'$ is defined likewise, so the bound of Section III prevents first-order cancellation.

### G. Feature images and normalization

The signed vector $x$ carries the security decision. For visualization and for the quantum encoding of Section VII we use nonnegative feature images derived from the PTM. The primary one is the cross-weight magnitude image $S^{\mathrm{cross,anti}}$, obtained by placing $|A_{\alpha\beta}|$ in both cross-weight orientations and zero elsewhere. Three complementary images capture other channel properties:

$$S_{\alpha\beta}^{\mathrm{anti,tr}} = \tfrac{1}{2}\,|R_{\alpha\beta} - R_{\beta\alpha}|\,, \qquad P_\alpha, P_\beta \neq I^{\otimes n}, \tag{8}$$

$$S_{\alpha}^{\mathrm{nonunital}} = |R_{\alpha,0}|, \qquad P_\alpha \neq I^{\otimes n}, \tag{9}$$

$$S_{\alpha\beta}^{\mathrm{cross,dev}} = |R_{\alpha\beta} - \delta_{\alpha\beta}|\,, \quad \{\mathrm{wt}(P_\alpha), \mathrm{wt}(P_\beta)\} = \{1, 2\}. \tag{10}$$

The antisymmetric-traceless image isolates coherent mixing over the whole traceless sector; the nonunital image isolates identity-to-Pauli leakage such as amplitude damping; the cross-weight deviation image collects all weight mixing, coherent or not. We also use the total deviation $S_{\alpha\beta}^{\mathrm{dev}} = |R_{\alpha\beta} - \delta_{\alpha\beta}|$ as a magnitude reference.

The cross-weight masks are symmetric in the two orientations because nonunital channels leak directionally: product amplitude damping turns a spectator identity into a $Z$ component, so a weight-1 column acquires weight-2 rows. A directed mask would report amplitude-damping cross mass 0, whereas the symmetric mask reports $\|S^{\mathrm{cross,dev}}\|_F = 0.064$ at matched infidelity 0.02; all results use the symmetric definition.

Each image is normalized in two stages: $\|S\|_F$ measures the amount of structure, $\widehat{S} = S/\|S\|_F$ retains its pattern, and FRQI loading uses $\widetilde{S}_j = S_j / \max_k S_k \in [0, 1]$; the local unital null $S = 0$ is reported directly as the absence of a cross-weight image.

For continuity with matched-infidelity sweeps we also report the normalized ratio

$$r_{\mathrm{cross,anti}} = \frac{\|S^{\mathrm{cross,anti}}\|_F}{\|S_{w=1}^{\mathrm{dev}}\|_F + \|S_{w=2}^{\mathrm{dev}}\|_F + \epsilon_0}, \tag{11}$$

where the denominator collects same-weight deviation mass and $\epsilon_0 > 0$ guards the zero denominator. The security statements are statements about $x$; the ratio is a reporting convention. For a pure attack channel every first-order low-weight entry is cross-weight, so the denominator is $O(\lambda^2)$ and the ratio admits the closed form

$$r_{\mathrm{cross,anti}}^{\mathrm{pure}}(\lambda) = \frac{2\sqrt{2}\,\sin\lambda}{6\,(1 - \cos\lambda)} = \frac{\sqrt{2}}{3}\cot(\lambda/2), \tag{12}$$

verified numerically to $10^{-6}$ over $\lambda \in [0.02, 0.8]$ and equal to 3.105 at matched infidelity 0.02.

## V. The Operational Detector

The defender deploys a single algorithm on the signed vector $x \in \mathbb{R}^{243}$ of the residual PTM.

1) *Estimate.* Acquire one shared randomized-Pauli classical-shadow dataset of $M$ total settings and form the estimate $\hat{x}$ from it. All entries read the same shots, so their estimators are correlated (Section VI-D).
2) *Calibrate.* From an attack-free window learn per-coordinate null means $\mu_j$ and standard deviations $\sigma_j$, regularized as $\sigma_j \leftarrow \max(\sigma_j, \kappa\,\mathrm{median}_j\,\sigma_j)$. We

fix $\kappa = 0.25$ before attack evaluation in all experiments so that near-zero-variance coordinates cannot dominate the maximum.

3) *Score.* Compute the direction-resolved statistic

$$T = \max_j \frac{|\hat{x}_j - \mu_j|}{\sigma_j}. \tag{13}$$

4) *Threshold.* Set $\tau$ to the $(1-\alpha)$ quantile of $T$ over a held-out attack-free window and flag if $T > \tau$.
5) *Corroborate.* Temporal-differential and context-contrast handles (Section VI-C) serve as secondary confirmation.

The max form is chosen over the plain norm $\|\hat{x} - \mu\|_2$ because the isometry of Section IV-E makes cross-weight directions faithful and non-cancelling, so a per-coordinate test can separate an attack direction from a native-coupling direction. Calibrating $\tau$ on the empirical distribution of $T$ absorbs the multiplicity of the maximum over 243 correlated coordinates (mean off-diagonal correlation 0.073, up to 0.38): an independence-based threshold gives false-positive rate 0.15 at target $\alpha = 0.05$, whereas the empirical quantile gives a held-out rate of 0.013 (95% Wilson interval 0.004–0.047) on 150 independent null records at $M = 8{,}192$.

The detector must not fire when benign drift changes between calibration and evaluation. For unital product noise with local drift, $x = 0$ throughout (Proposition 1); for nonunital product noise, the drift PTM acts orthogonally on the cross-weight block, so $\|x\|_2$ is exactly invariant while the vector rotates by $O(\epsilon)$. Numerically, the vector changes by at most $10^{-16}$ over $\epsilon \in [0, 0.3]$ in the unital case, and under realistic noise the norm changes by $3 \times 10^{-17}$ while the vector moves by about $10^{-4}$, so the calibration window should span the expected drift range.

## VI. Evaluation

All experiments use $n = 3$ qubits and the $k = 2$ sector ($K = 37$), simulated with custom Python 3.10 code using NumPy and SciPy, without Qiskit or QuTiP: PTMs are built from Kraus operators, pulses use time-ordered integration with 400 steps per gate, and FRQI/QHED is a 13-qubit statevector. Realistic noise is thermal relaxation with $T_1 = 50$ $\mu$s and $T_2 = 70$ $\mu$s over a 200-ns gate, depolarizing probability $10^{-3}$, and amplitude damping, composed as $\mathcal{E} = \mathcal{N}_{T_1,T_2,p} \circ \mathcal{A}_\lambda \circ \mathcal{M}_\epsilon$. Unless noted, compared families are matched by numerical bisection in average gate infidelity $r = 1 - F_{\text{avg}}$, with $F_{\text{avg}} = (\text{Tr}R + d)/(d(d+1))$ and $d = 2^n$, and finite-shot studies use fixed seeds.

The finite-shot studies follow a common protocol. Null records used to estimate $\mu_j$ and $\sigma_j$ are disjoint from those used to select thresholds, and reported false-positive rates use a further held-out set, preventing optimistic reuse of the null samples. The detection-limit study of Section VI-D uses moment, threshold, and per-coupling attack record counts that decrease from 240/240/140 at $M = 4{,}096$ to 60/60/35 at $M = 16{,}384$; the held-out false-positive rate of the same detector is reported in Section V. The native ZZ study of Section VI-E uses 200 records each for moments, threshold selection, and held-out reporting, with 120 attack records per coupling. Every record is scored from a fresh shared-shadow dataset of $M$ single-shot settings.

TABLE II

Feature Frobenius norms at matched average gate infidelity 0.02 for $n = 3$, $k = 2$. The coherent rotation dominates the antisymmetric-traceless feature, while amplitude damping dominates the nonunital and cross-weight deviation features.

| Channel | $\|S^{\text{anti,tr}}\|_F$ | $\|S^{\text{nonunital}}\|_F$ | $\|S^{\text{dev}}\|_F$ | $\|S^{\text{cross,dev}}\|_F$ |
|---|---|---|---|---|
| $R_x$ rotation | 1.1104 | 0.0000 | 1.1231 | 0.0000 |
| Depolarizing | 0.0000 | 0.0000 | 0.1084 | 0.0000 |
| Dephasing | 0.0000 | 0.0000 | 0.1225 | 0.0000 |
| Amp. damping | 0.0450 | 0.0262 | 0.1318 | 0.0636 |

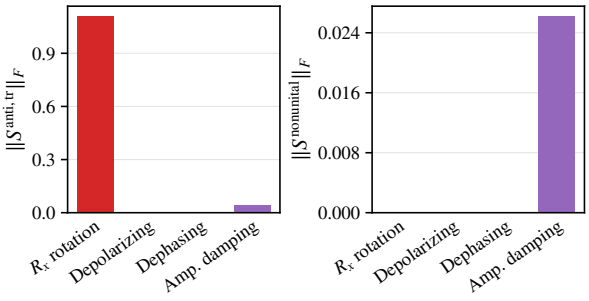



Fig. 2. Feature norms for four channel families at matched average gate infidelity 0.02. The antisymmetric-traceless feature separates coherent rotation from the tested incoherent channels, while the nonunital feature identifies amplitude damping.

### A. Mechanism separation at matched infidelity

We first verify that the feature images of Section IV-G respond to the mechanism of an error rather than its magnitude. Table II and Fig. 2 compare a local coherent rotation, depolarizing noise, dephasing, and amplitude damping at average gate infidelity 0.02: the antisymmetric-traceless norm isolates the coherent channel with a margin of $+1.065$, and the nonunital norm isolates amplitude damping, neither being visible to the common scalar fidelity.

### B. Drift versus attack: the structural separation (RQ1)

The central experiment matches local coherent drift against ZX, YX, and ZZ attacks at average gate infidelities from 0.005 to 0.06. In Fig. 3 (right), the drift value of $r_{\text{cross,anti}}$ is identically zero, as Proposition 1 requires, while all attacks follow Eq. (12); at infidelity 0.02 the attack value is 3.105, a zero-versus-nonzero distinction. This answers RQ1 in the noiseless limit; the rest of this section addresses RQ3.

The figure also retains a failed baseline for contrast. The weight-sector ratio $r_w$, the edge score of the weight-2 block over the weight-1 block, fails because local drift on one qubit also produces weight-2 tensor copies: at matched infidelity 0.02 the drift value $r_w = 13.0$ exceeds the attack value $r_w = 8.0$ (Fig. 3, left). Scalar infidelity and RB error rates do not locate the coherent component, $r_w$ confounds product and interaction structure, the scalar cross-weight norm loses

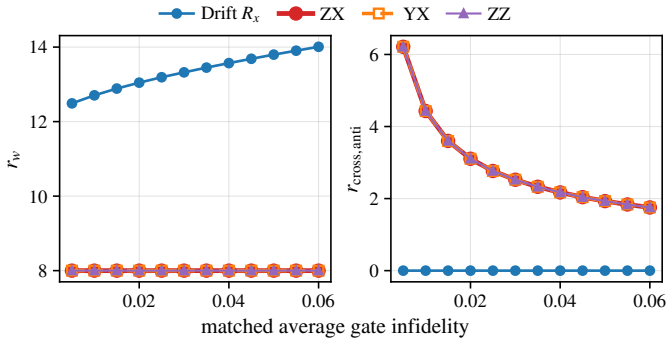



Fig. 3. Left: the weight-sector ratio $r_w$ fails; local drift exceeds all attacks at matched infidelity because product drift creates weight-2 copies. Right: the corrected discriminator $r_{\mathrm{cross,anti}}$ separates exactly; drift is identically zero (Proposition 1) while all three attack families follow $r^{\mathrm{pure}}_{\mathrm{cross,anti}} = \frac{\sqrt{2}}{3}\cot(\lambda/2)$ over the tested range. This normalized ratio is used only to visualize mechanism separation; it is not the operational detection statistic, which is the calibrated test of Eq. (13) on the signed vector $x$.

directionality, and the direction-resolved statistic of Eq. (13) retains the interaction geometry.

Real devices are not products of unital channels, and amplitude damping gives a small nonzero raw value ($r_{\mathrm{cross,anti}} = 0.016$ with no attack under the full noise model).

Baseline subtraction against a calibration estimate of the same device restores the separation: the no-attack differential returns to zero while ZX, YX, and ZZ attacks retain $\Delta r_{\mathrm{cross,anti}} \approx 1.05$ at infidelity 0.02. Under shadow noise at 65,536 settings (200 trials), the separation stays clean, with attack minimum 0.88 against a drift and noise holdout maximum of 0.17.

### C. Temporal and context handles

Two secondary handles, specializations of the context and drift tests of [15], corroborate an alarm. The temporal differential scores $\|S_t - \bar{S}\|_F$ against a calibration-window mean: slow drift moves it smoothly, while attack insertion produces a jump. In a 12-frame timeline with an attack inserted at frame 8, the first attack frame scores 0.91 against drift frames bounded by 0.30 (Fig. 4).

The context contrast compares residual features with the spectator idle versus driven. Victim-local drift commutes with a spectator-only drive, so its contrast is exactly zero, while two-qubit attack terms do not commute and produce a contrast that grows with infidelity (Fig. 5). The handle is scoped to victim-local drift; spectator-local drift is already removed by the cross-weight signature.

### D. Finite measurements and detection limits (RQ3)

The low-weight block is estimated from a single shared dataset, as on hardware and in randomized product-Pauli characterization [6]. Each setting draws one random product-Pauli input eigenstate and one random measurement basis, is executed with a single shot, and contributes to every PTM entry it covers; $R_{\alpha\beta}$ is covered when the bases match the supports of $P_\alpha$ and $P_\beta$, and the two-sided inverse-shadow

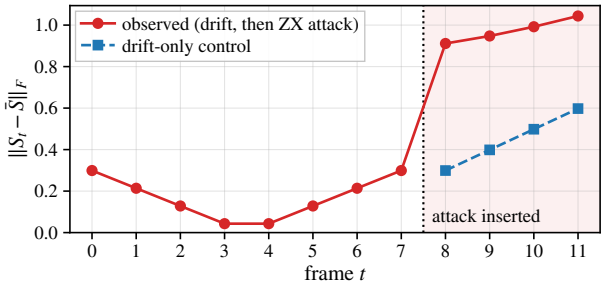



Fig. 4. Temporal differential over a 12-frame timeline with attack insertion at frame 8. Drift frames stay below 0.30; the first attack frame scores 0.91.

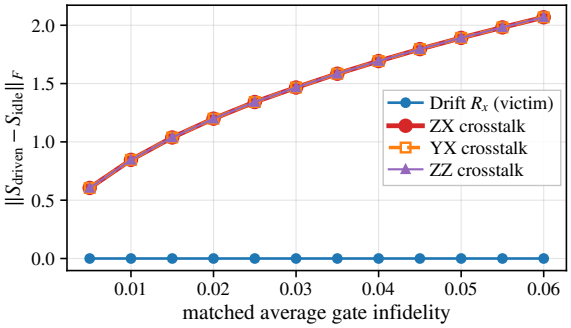



Fig. 5. Context contrast between driven and idle spectator conditions across matched infidelities. Victim-local drift cancels exactly; all three crosstalk families produce nonzero contrast at every tested infidelity.

factor $3^{w_\alpha + w_\beta}$ cancels the match probability in expectation. Budgets $M$ count total settings, equal here to total shots, and measure sample complexity rather than audit latency; the per-entry cross-weight variance scales as $27/M$, so the noise floor on $\hat{x}$ scales as $M^{-1/2}$.

Figure 6 reports the resulting detection limit $\lambda_{\min}$ of the detector of Section V, the smallest coupling reaching true-positive rate at least 0.95 at the threshold calibrated for $\alpha = 0.05$, as a function of $M$. The fitted exponents are $-0.51$ for ZX, $-0.56$ for YX, and $-0.52$ for ZZ, all consistent with the predicted $M^{-1/2}$ scaling, and the three families track one another, as expected from the isometry of Lemma 1. At $M = 16{,}384$ total settings the limit is $\lambda_{\min} \approx 0.13$ for all families under the two-sided accounting. The tested couplings increase by a factor of about 1.5, and $\lambda_{\min}$ is log-interpolated between the two that bracket the 0.95 crossing, so the limits and exponents are resolved only to that bracket.

### E. The native ZZ baseline (RQ3)

Motivated by transmon characterization [3], the calibration channel carries an always-on ZZ coupling $\zeta_{\mathrm{cal}} = 0.10$ rad per 200-ns gate that fluctuates as $\zeta_{\mathrm{eval}} = \zeta_{\mathrm{cal}} + \delta\zeta$, $\delta\zeta \sim \mathcal{N}(0, \sigma^2_\zeta)$. The null is native wander only; the attack adds excess coupling on the same pair. All channels pass through the two-sided estimator at 65,536 settings, and we compare the scalar norm with the direction-resolved statistic of Eq. (13).

Figure 7 shows that the resulting hardware floor is directional, as the geometry of Section IV-E predicts. Excess

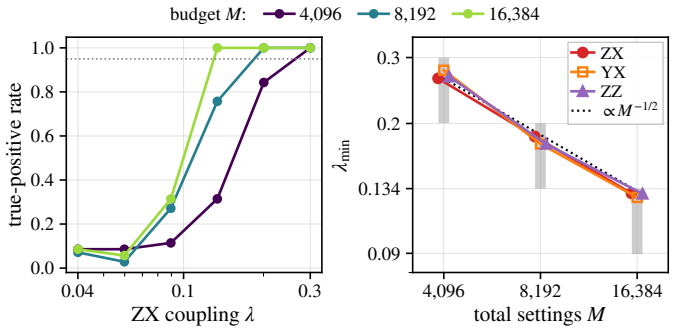



Fig. 6. Detection limit of the calibrated maximum-$z$ detector under two-sided shared-dataset shadow estimation. Left: ZX true-positive rate versus coupling for increasing total budgets $M$. Right: $\lambda_{\min}$ versus $M$ for all three families (markers offset horizontally for visibility) with the $M^{-1/2}$ guide; gray bars mark the adjacent tested couplings that bracket the 0.95 crossing.

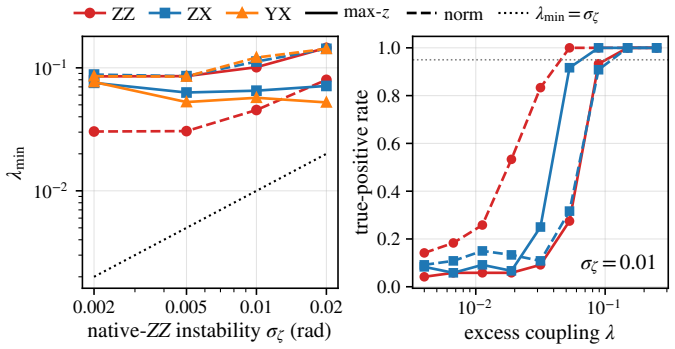



Fig. 7. Calibrated native ZZ baseline ($\zeta_{\text{cal}} = 0.10$, two-sided shadows, 65,536 settings). Left: $\lambda_{\min}$ versus instability $\sigma_\zeta$ for excess ZZ, ZX, and YX under the scalar norm and the direction-resolved statistic. Right: true-positive rate at $\sigma_\zeta = 0.01$. Excess ZZ is floor-limited; ZX and YX under the direction-resolved statistic stay at the shot floor.

ZZ points along the native-wander direction, so its limit rises with instability regardless of statistic, from the shot floor 0.030 at $\sigma_\zeta = 0.002$ to about $0.080 \approx 4\sigma_\zeta$ at $\sigma_\zeta = 0.02$. Attacks along other directions populate cross-weight coordinates that native wander never touches: with the direction-resolved statistic, ZX and YX hold $\lambda_{\min} \approx 0.05$ to 0.08 essentially independent of $\sigma_\zeta$, while the scalar norm degrades by about a factor of two. Tracking the native coupling [3] recovers part of the collinear component.

Real calibration drift can also be heavy-tailed, temporally correlated, or dominated by discrete two-level-fluctuator jumps. Repeating the experiment with the null drawn from a Student-$t$ distribution (three degrees of freedom), an AR(1) process ($\rho = 0.8$), and symmetric telegraph jumps, all matched to $\sigma_\zeta = 0.04$ at $M = 16{,}384$, preserves the directional advantage: off-collinear ZX limits stay in $[0.11, 0.16]$ near the shot floor, while collinear ZZ limits remain instability-limited in $[0.18, 0.30]$. The hardest case is temporally correlated drift, which narrows the gap and pushes the held-out false-positive rate to 0.10 against a target of 0.05, because a colored null is poorly summarized by per-coordinate moments; strongly autocorrelated nulls require block calibration, an effective sample-size correction, or a time-series-aware threshold.

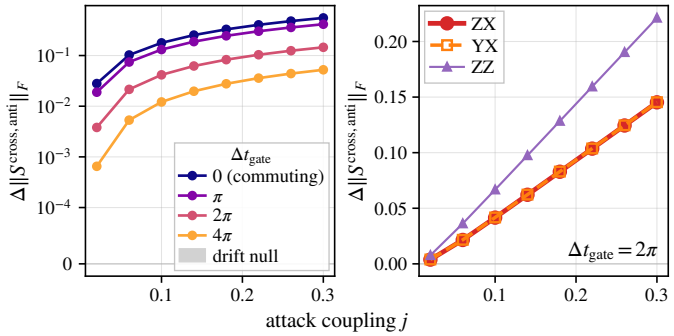



Fig. 8. Time-ordering-sensitive pulse validation. Left: numerator differential versus bare ZX coupling $j$ for detunings $\Delta t_{\text{gate}} \in \{0, \pi, 2\pi, 4\pi\}$; the shaded band is the drift-varied null, which is machine zero for the numerator. Right: all three attack families at $\Delta t_{\text{gate}} = 2\pi$. Detection persists under genuinely non-commuting evolution.

### F. Pulse-level validation (RQ3)

Constant-generator unitaries are algebraic stand-ins for real drives, so we validate the detector on time-dependent models. A Hamiltonian residual model factors out an intended drive $H_{\text{id}} = \frac{\pi}{4} X_v + \frac{\pi}{6} X_s$ before scoring, with realistic noise; ZX, YX, and ZZ terms give $\Delta r_{\text{cross,anti}} = 0.110$, 0.109, and 0.097 at 0.02 rad, growing monotonically to 1.61, 1.61, and 1.41 at 0.30 rad. A cross-resonance pulse whose terms share one Gaussian-square envelope reproduces these values to three figures, as it must, since a common envelope commutes with itself; this checks the pipeline, and the integrator matches the exact exponential to $2.6 \times 10^{-14}$.

The third model makes time ordering nontrivial: the attack term carries its own detuned, phase-modulated envelope, $H(t) = \Omega(t) H_{\text{drive}} + \frac{j}{2}\Omega(t)\cos(\Delta t + \varphi)\, G$, with $\Delta t_{\text{gate}} \in \{\pi, 2\pi, 4\pi\}$ and $\varphi = \pi/4$. The witness $\|U_{\text{pulse}} - e^{-i\int H}\|$ reaches 0.06, against $10^{-14}$ for the common envelope, so the evolution departs from the un-ordered integrated Hamiltonian. Two results follow (Fig. 8). First, for drift-varied nulls with $\epsilon_{\text{eval}} \in [0.04, 0.12]$ around the calibration value 0.08, the numerator differential $\Delta\|S^{\text{cross,anti}}\|_F$ stays at $4 \times 10^{-16}$, the empirical face of the drift invariance of Section V, while the ratio moves by 0.028 through its denominator. Second, detection persists at every detuning: the envelope overlap $\chi(\Delta) = |\int \Omega \cos(\Delta t + \varphi)\, dt| / \int \Omega\, dt$ falls from 0.71 to 0.07 at $\Delta t_{\text{gate}} = 4\pi$, and all curves remain monotone in $j$ and clear the drift-null band at the smallest tested coupling.

### G. Resource scaling

Table III summarizes growth at fixed $k = 2$, with counts for $n \leq 4$ taken from the exact graph builder and larger $n$ from closed forms validated against it. The monitored objects grow polynomially: $K = \Theta(n^2)$ low-weight Paulis, $\Theta(n^3)$ cross-weight coordinates, and $\Theta(n^4)$ generic FRQI controlled rotations, with an address register of only $\Theta(\log n)$ qubits. Each cross-weight coordinate has a two-sided shadow variance factor of at most $3^{2k} = 81$, constant at fixed $k$. Because

TABLE III
RESOURCE SCALING AT FIXED $k = 2$. COUNTS FOR $n \leq 4$ ARE EXACT; $n \geq 5$ FROM CLOSED FORMS VALIDATED AT $n \leq 4$.

| $n$ | $K$ | Cross entries | Hamming edges | FRQI qubits | FRQI rot. |
|---|---|---|---|---|---|
| 3 | 37 | 243 | 9 324 | 11 | 1 369 |
| 4 | 67 | 648 | 32 160 | 13 | 4 489 |
| 5 | 106 | 1 350 | 82 680 | 14 | 11 236 |
| 6 | 154 | 2 430 | 177 408 | 15 | 23 716 |

one shared dataset estimates all coordinates simultaneously, the total budget is not a per-entry cost multiplied by the entry count; the entry count controls classical storage and post-processing.

Beyond three qubits, commuting Ising residuals $U = \exp[-i \sum_{(i,j)\in E} \theta_{ij} Z_i Z_j/2]$ admit the exact response

$$R_{Y_i Z_j, X_i} = \sin\theta_{ij} \prod_{k \in N(i)\setminus\{j\}} \cos\theta_{ik}, \tag{14}$$

where $N(i)$ denotes the neighbors of qubit $i$. Thus cross-weight visibility depends on interaction angles and neighborhood degree. Pairwise full-block norms $b_{ij}$, restricted to weight-1 and weight-2 Paulis supported on $\{i, j\}$, retain invariance under local-unitary pre- and postcomposition because product unitaries preserve Pauli support, providing features for graph-based analysis of distributed coupling anomalies. FRQI/QHED can implement the corresponding graph readout; larger-system finite-shot performance, localization accuracy, and end-to-end quantum advantage remain to be evaluated.

## VII. QUANTUM-NATIVE PAULI-GRAPH EDGE READOUT

The detector operates on the signed vector $x$. RQ4 asks whether the PTM-derived image admits a quantum-native representation and readout; this section constructs one and states what it does and does not provide.

### A. FRQI encoding of the PTM feature

Let $\widetilde{S}_j \in [0, 1]$ denote the normalized nonnegative feature values, padded with zeros to $N_{\text{pad}} = 2^m$ entries. The FRQI state is

$$|\text{FRQI}(S)\rangle = \frac{1}{\sqrt{N_{\text{pad}}}} \sum_{j=0}^{N_{\text{pad}}-1} |j\rangle \left( \sqrt{1 - \widetilde{S}_j^2}\, |0\rangle + \widetilde{S}_j |1\rangle \right), \tag{15}$$

prepared by controlled rotations with angles $\theta_j = \arcsin \widetilde{S}_j$ [11]; the address register identifies a PTM coordinate and the final qubit stores its magnitude. Postselecting the final qubit on $|1\rangle$ yields $|\psi_S\rangle = \|S\|_2^{-1} \sum_j S_j |j\rangle$ with success probability $\sum_j \widetilde{S}_j^2 / N_{\text{pad}}$, which vanishes exactly for the local unital null. The state keeps magnitude and support but not sign, so $x$ remains the detection object.

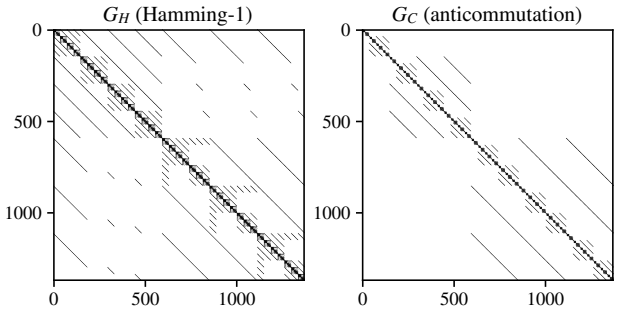



Fig. 9. Adjacency of the two Pauli graphs on the $K \times K$ low-weight block at $n = 3$, $k = 2$ ($K = 37$). Left: Hamming-1 graph, 9,324 edges. Right: anticommutation graph, 4,662 edges.

### B. Pauli graph and QHED score

A PTM pixel is indexed by a pair of Pauli strings $(P_\alpha, P_\beta)$, and lexicographic adjacency between such pairs is physically arbitrary, so the cyclic-shift neighborhood of standard QHED [12] does not apply. Our Pauli graph $G = (V, E)$ connects PTM coordinates in which one label is unchanged and the other differs in one tensor factor; a sparser variant also requires the differing factors to anticommute. At $n = 3$, $k = 2$ the two graphs have 9,324 and 4,662 edges in 18 and 9 matchings (Fig. 9).

The normalized Pauli-graph edge score is the graph Dirichlet energy

$$\mathcal{A}^G_{\text{edge}}(S) = \frac{1}{4\sum_j S_j^2} \sum_{(u,v)\in E} |S_u - S_v|^2 = \frac{S^T L_G S}{4\, S^T S}, \tag{16}$$

where $L_G$ is the graph Laplacian. QHED implements it through the matching decomposition: each matching $M_\ell$ defines an involutive permutation $D_{M_\ell}$ of the address register, and a Hadamard on an auxiliary qubit, a controlled $D_{M_\ell}$, and a second Hadamard yield $P_\ell = \frac{1}{4}\sum_j |a_j - a_{D_{M_\ell}(j)}|^2$ for $a_j = S_j/\|S\|_2$, so $\mathcal{A}^G_{\text{edge}}(S) = \frac{1}{2}\sum_\ell P_\ell$.

The detector and the QHED score answer different questions: $\|x\|_2$ measures the amount of cross-weight coherent structure, while $\mathcal{A}^G_{\text{edge}}(S)$ measures how that structure is arranged over the Pauli graph.

### C. Circuit verification and ablation (RQ4)

A gate-level statevector implementation with 11 address qubits, one pixel qubit, and one auxiliary qubit executes FRQI loading, postselection, and the edge primitive on every matching of both graphs, and reproduces the classical edge score for all tested channels, features, and graphs with maximum error $1.4 \times 10^{-14}$ (Fig. 10), which answers RQ4 at the level of exact realization.

Two qualifications bound the claim. First, the implemented route is classically compiled: the PTM is estimated classically, the FRQI rotations are compiled from those values, and the graph score is classically computable in $O(|E|)$ time once $S$ is known, so no quantum speedup is claimed; a fully coherent route needs a PTM-derived value oracle or equivalent

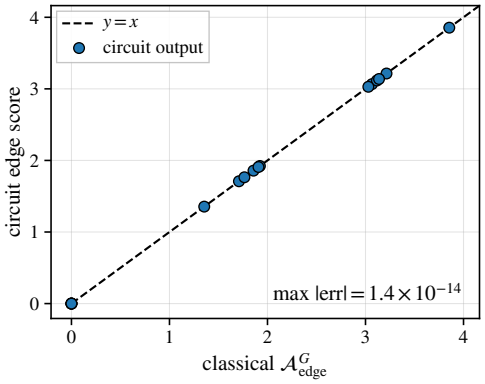



Fig. 10. Circuit-level verification of the Pauli-graph QHED readout: the statevector circuit matches the classical edge score across all tested channels, features, and both graphs (maximum error $1.4 \times 10^{-14}$).

TABLE IV
ABLATION: DRIFT-VERSUS-ZX DETECTION ROC-AUC UNDER SHADOW NOISE. STRUCTURAL STATISTICS DOMINATE; QHED SERVES AS THE QUANTUM-NATIVE MORPHOLOGY READOUT.

| $\lambda$ | $\|x\|_2$ | max-$z$ | graph-TV | QHED edge |
|---|---|---|---|---|
| 0.05 | 0.599 | 0.542 | 0.571 | 0.551 |
| 0.10 | 0.748 | **0.787** | 0.679 | 0.662 |
| 0.15 | 0.936 | **0.979** | 0.852 | 0.882 |

block encoding. Second, QHED is not the strongest detector: in the drift-versus-ZX ablation of Table IV, the structural statistics dominate at every coupling, while QHED remains the quantum-native readout of graph morphology.

FRQI/QHED provides a quantum readout of graph-structured coupling features. Scaling benefits could arise when coherent feature preparation avoids explicit classical reconstruction, or when reversible access enables amplitude amplification and estimation; an end-to-end advantage must account for feature acquisition, state preparation, graph operations, and competing classical sampling methods. For a unitary residual $U$ with $U$ and $U^\dagger$ applicable to the monitored qubits, or equivalently $U$ and $U^*$ on the halves of a Bell register $|\Phi\rangle$, Pauli-Bell encoding of $(UP_\beta U^\dagger \otimes I)|\Phi\rangle$ over the $3n$ weight-1 inputs, Bell decoding, and postselection on weight-2 outputs prepare the normalized signed $R_{21}$ state $\propto \sum_{\alpha,\beta} R_{\alpha\beta}|\alpha\rangle|\beta\rangle$ with success probability $\|R_{21}\|_F^2/(3n)$. For pure Ising dynamics on a bipartite coupling graph, $X$ echo pulses on one partition implement $U^\dagger = U^*$. This gives a concrete route to QHED without classical feature loading; robustness under noisy dynamics and an end-to-end advantage remain to be established.

## VIII. Limitations and Countermeasures

*a) Local-rotation evasion:* The evaluated statistic vanishes for an attack composed with a local $\pi$ rotation (Section IV-F); the full-block norm removes this cancellation but still requires finite-shot calibration.

*b) Incoherent attacks:* A stochastic correlated injection, such as random-sign ZZ or correlated dephasing, produces a Pauli-diagonal PTM with zero cross-weight mass and evades the discriminator, a boundary inherent to any coherence-based detector. Magnitude features and the temporal handle give partial coverage, and purity-based or randomized benchmarking diagnostics target this class.

*c) Weight truncation:* Coherent generators of weight greater than $k$ escape the monitored sector. Raising $k$ trades polynomially more estimation cost for coverage.

*d) Native-crosstalk mimicry:* Excess ZZ within the native fluctuation band is indistinguishable from calibration wander, with $\lambda_{\min} \approx 4\sigma_\zeta$. The floor is directional, so non-collinear attacks remain detectable at the shot floor, and tracking the native coupling [3] tightens the collinear floor.

*e) Baseline poisoning:* All differential handles assume an attack-free calibration window. An adversary active during calibration is absorbed into the baseline. Randomizing calibration scheduling mitigates but does not eliminate this.

*f) Diagnostic-window evasion:* A context-aware adversary who attacks only tenant workloads and stays silent during audits is not observed. Provider-side deployment should interleave audits randomly within tenant execution so that audit and workload gates are statistically indistinguishable to the attacker; we do not analyze the resulting game, the most important open adversarial question.

## IX. Conclusion

This work establishes a structural approach to verifying multi-tenant quantum execution beyond scalar error metrics. For RQ1, products of local unital errors yield an exactly zero cross-weight PTM block, while every nonzero two-qubit interaction contributes at first order. For RQ2, the map from a specified pair's nine interaction coefficients to the antisymmetric feature is an isometry up to scale for every system size, ensuring sensitivity to all interaction directions near the identity. The full cross-weight norm preserves this signal under arbitrary local unitary composition, extending the structural guarantee beyond the evaluated detector's small-rotation regime. For RQ3, the calibrated detector uses one shared randomized-Pauli dataset to identify ZX, YX, and ZZ interactions, reaching $\lambda_{\min} \approx 0.13$ at 16,384 settings with a 5% calibrated false-positive target and approximately $M^{-1/2}$ scaling. Experiments support detection under relaxation, depolarization, and pulse dynamics, while native ZZ fluctuations reveal directional detection floors and colored drift motivates improved temporal calibration. For RQ4, QHED on the FRQI-encoded feature reproduces the classical Pauli-graph edge score within $1.4 \times 10^{-14}$; coherent feature preparation remains necessary for an inline quantum diagnostic.

Together, these results connect exact structural guarantees to measurement-based detection. Operator-indexed features derived from basis algebra offer a broader route to quantum verification, with device validation using measured drift and injected pulse perturbations as the next step.

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